

BCG Data Science – Forage presentation

~By Ankita Sarkar



Task-1

To ensure we can successfully identify the cause, your task is to draft an email clearly outlining the data that we'll need from the client and the techniques you will use to investigate the issue. To help get you started, ask yourself:

What are all of the possible reasons a customer would choose to stay with or switch energy providers? (for example: price, energy source, customer service, etc.)

What data points would be needed in order to effectively analyze causes of churn? (e.g. purchasing trends, location of business etc.)

Once we have the data, what steps will you take to process and analyze the information? (hint: use the 5-step process outlined previously)

My Email: Data Required & Analytical Approach

Hi Estelle,

Following our discussion with the client, I have outlined the data we would need and the analytical approach I propose to investigate the hypothesis that customers are becoming more price-sensitive and switching to lower-cost energy providers.

1. Business understanding and hypotheses

Although price may be an important driver of churn, customers may switch providers for several other reasons, including:

- Energy prices and perceived value for money

- Contract terms, renewal dates, and early termination fees

- Customer service quality and satisfaction

- Reliability of energy supply and service

- Availability of renewable/green energy options

- Billing accuracy and payment experience

- Discounts, promotions, and competitor offers

- Customer location and availability of alternative providers

- Changes in customer energy consumption or business circumstances

- Length of relationship with the provider

Therefore, we should test price sensitivity alongside these other potential factors rather than assuming price is the primary cause.

2. Data required from the client

To investigate these factors, I would request:

- Customer information: customer ID, location, customer type/segment, acquisition date, and tenure
- Churn information: churn status, churn date, reason for leaving, and previous churn history
- Pricing data: current and historical energy prices, tariffs, discounts, and fees paid by each customer
- Contract data: contract type, start/end dates, renewal dates, and termination fees
- Usage data: historical energy consumption, monthly usage patterns, and changes in consumption
- Customer service data: number of complaints, support interactions, resolution times, and satisfaction scores
- Competitor data: competitor prices, available plans, promotions, and providers available in the customer's location
- Payment/billing data: billing amounts, payment history, late payments, and billing issues
- Energy preferences: renewable-energy usage/preferences where available
- Customer feedback: survey responses or recorded churn reasons

Having historical data would be particularly important so that we can compare customer behaviour before churn with that of customers who remained.

3. Data preparation and exploration

Once the data is received, I would first validate its quality by checking for missing values, duplicates, inconsistent formats, outliers, and incorrect records. I would then combine the relevant datasets using a unique customer identifier and create useful variables such as price changes, percentage price differences, customer tenure, usage changes, and complaint frequency.

I would then perform exploratory data analysis to identify patterns in churn across customer segments, locations, contract types, price levels, and usage behaviour.

4. Analysis and testing

To specifically test the client's price-sensitivity hypothesis, I would:

- Compare churn rates between customers exposed to different price levels and price increases.

- Analyse whether customers who experienced larger price increases were more likely to churn.

- Compare the prices paid by churned customers with those of retained customers.

- Examine whether churn increases when competitors offer lower prices.

- Segment customers by factors such as tenure, location, contract type, and usage to determine whether price sensitivity differs across groups.

- Use correlation analysis and statistical testing to identify meaningful relationships.

- Build a logistic regression or similar classification model to estimate the probability of churn and measure the contribution of price while controlling for other factors such as service issues, contract type, tenure, and customer characteristics.

- This will help distinguish whether price is genuinely associated with churn or whether other factors explain the observed relationship.

5. Communicate findings and recommendations

Finally, I would summarise the key drivers of churn and assess the evidence for the price-sensitivity hypothesis. I would present the results using clear visualisations and explain which factors have the strongest relationship with churn, including any important differences between customer segments.

If the analysis shows that price is a significant driver, the client could investigate targeted pricing, discounts, or retention offers. If other factors have a stronger relationship with churn, the findings could instead guide improvements in areas such as customer service, contracts, or billing.

This approach should allow us to move from the initial hypothesis to a data-driven understanding of why customers are leaving and provide the client with evidence to support their retention strategy.

Best regards,

Ankita

Thank You

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